DiaTwin

A Type 2 diabetes digital twin that fuses EHR and wearable data

Happiest Health Digital Twin Challenge 2026

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Educational prototype. Trained and evaluated on synthetic data. Not a medical device.

Problem and healthcare use case

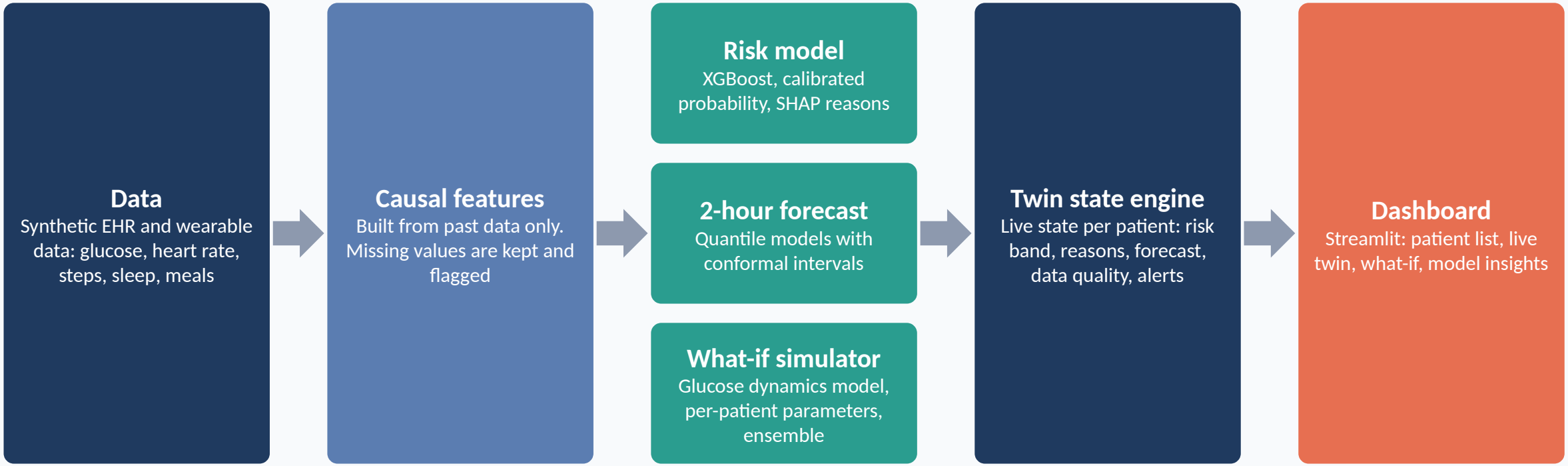
Problem

- Glucose monitors report a rise after it has happened.
- A care team needs a warning before a post-meal rise, with reasons, and a way to compare meal and activity options.
- Health records and wearable streams are rarely combined into one patient model.

Use case

- A dashboard for a Type 2 diabetes care team, and a learning tool.
- Risk of a glucose rise of 50 mg/dL or more in the next 2 hours, with plain-language reasons.
- A 2-hour glucose forecast with a range.
- What-if scenarios for meals and activity only. No insulin or medication scenarios.

Architecture



Frozen artifacts with SHA-256 manifests | patient-wise splits | causal features | test evaluated after the freeze

Data and evaluation protocol

Synthetic cohort

- 40 synthetic patients, 21 days each, readings every 5 minutes.
- Glucose, heart rate, steps, sleep, meals and an EHR profile.
- Patient-wise train, validation and test split (28, 6, 6).
- Features use past data only.

Real cohort (realism check)

- ShanghaiT2DM: 100 patients, 109 recordings, 112,462 CGM readings.
- Mean glucose about 141 mg/dL (real) against about 219 mg/dL (synthetic), patient-level.
- The models are not validated on real people.

Integrity record

- Model, calibrator and thresholds are frozen with SHA-256 manifests.
- Risk model: the freeze commit precedes the test evaluation in git.
- Forecast and what-if: manifest and test marker were committed together, so the order cannot be shown from git.

Results in this synthetic setting

Risk model (test, 6 patients)

- ROC-AUC 0.925 (95% CI 0.907 to 0.944).
- Calibration error (ECE) 0.0275.
- High band: precision 61.0%, recall 67.2% (row level).
- Medium band: precision 41.0%, recall 91.4% (row level).

Forecast (validation)

- 60-minute RMSE 18.41 mg/dL.
- The 80% interval covers 80.0% on the split used to calibrate it.
- Held-out test metrics were not saved.

What-if (stored test result)

- Peak error 1.0, 10.2 and 11.5 mg/dL for three scenarios.
- 80% effect intervals covered the true effect 0%, 17% and 0% of the time.
- The bands under-cover; read the ranges as optimistic.

Dashboard and example moment

Pages

- Patient List.
- Live Twin: risk band, top three reasons, 2-hour forecast band.
- What-if Simulator: meal and activity scenarios with a confidence label.
- Model Insights: metrics, integrity record, limitations.
- About, and an experimental page for your own readings (local only).

Example moment (synthetic)

- Patient P007 at 21:50.
- Glucose 191.4 mg/dL, risk band High.
- Calibrated probability 0.967.
- Predicted peak 302.3 mg/dL.
- A selected example, not a typical moment.

Limitations and next steps

Limitations

- Synthetic data and simplified physiology; synthetic glucose runs higher than the real cohort.
- Only 6 test patients, so the intervals are wide.
- What-if intervals under-cover; forecast test metrics were not saved.
- Not validated on real people. Type 2 diabetes only. No medication modelling.

Next steps

- Rebuild the synthetic cohort to match the real glucose level.
- Validate on real CGM data with patient-wise cross-validation.
- Calibrate the what-if intervals and tune the alert policy.
- Run a held-out evaluation for the forecast.